Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	McPherson, Alexander Ph.D.		
Project Title:	An Observable Protein Crystal Growth Flight Apparatus		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	amcphers@uci.edu	**Fax:**	FY 949-824-1954
PI Organization Type:	UNIVERSITY	**Phone:**	949-824-1931
Organization Name:	University of California, Irvine		
PI Address 1:	Department of Molecular Biology and Biochemistry		
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PI Web Page:	http://www.faculty.uci.edu/		
City:	Irvine	**State:**	CA
Zip Code:	92697-3900	**Congressional District:**	48
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	05/01/1993	**End Date:**	07/01/2004
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Segre, Philip N	**Contact Phone:**	256 544-1597
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Flight Program:			
Flight Assignment:	NOTE: During FY04 reporting period, the OPCGA was redirected from a flight program to a ground science program per information in Task Book report (Ed., 10/3/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Malkin, A. (University of California Irvine) Kuznetsov, Y. (University of California Irvine) Jenkins, G. (University of California Irvine) Wells, M. (University of California Irvine)		
Grant/Contract No.:	NAS8-00017		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The experiment objective includes the identification and characterization of candidate biochemical systems that include proteins, nucleic acids, and viruses. It further includes the construction of an optical platform that would be suitable for detailed interferometric analysis of protein crystal growth experiments and the visualization of concentration fields, the time lapse microphotography of macromolecular crystals, and the further characterization of the mechanisms and fundamental parameters that determine the features of macromolecular crystal growth. The task objective is to initiate research and development efforts in the area of macromolecular crystal growth, and specifically focused on the direct observation of the relevant phenomena as they pertain to the design and ultimate flight. The task objective is to initiate research and development efforts in the area of macromolecular crystal growth, and specifically focused on the direct observation of the relevant phenomena as they pertain to the design and ultimate flight of an observable protein crystal growth apparatus (OPCGA). The need for structural information on biological macromolecules is of paramount importance to the emerging field of biotechnology. Such information provides the basis for the rational design of pharmaceuticals, the determination of enzymatic mechanisms and the engineering of proteins to enhance or modify their function. The value of x-ray crystallography to provide structural information at atomic resolution is unsurpassed. This technique does, however, depend on the availability of crystals for the macromolecule under study, and furthermore, which possess a high degree of internal order and suitable size and shape to allow the accurate collection of x-ray diffraction data. Typically, crystals grown in the earth's gravitational field suffer from one or more types of flaws which decrease the structural information that can be derived from them. These imperfections include the relatively simple, but nonetheless adverse, problem of intergrowth. In a 1-g field several crystals, which nucleate independently, sediment to the bottom of the growth chamber where they intergrow into a mass not suitable for x-ray diffraction analysis. Such intergrowth or the appearance of satellite crystals, which were earlier thought to occur by nucleation at the sides of defects in pre-existing crystals, have been shown to be almost exclusively due to the effects of sedimentation and intergrowth. The value of a microgravity environment for the elimination of sedimentation in the preparation of these crystals is obvious. Early experiments conducted on board the Space Shuttle clearly indicated such benefits. They also provided the impetus for other researchers to design experiments and procedures for mimicking the effect of microgravity in ground-based experiments. The development of methods for successfully growing protein crystals in gels is an example of such activities.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During this reporting period, the OPCGA was redirected from a flight program to a ground science program. Efforts focused on disposition of residual hardware, completion of in-process hardware development and testing, and closure of in-process verification activities. The hardware has been prepared and stored in a manner to maintain the hardware flight pedigree until it is shipped back to the UCI Campus where it will be used to conduct ground based experiments.
Bibliography Type:	Description: (Last Updated: 02/27/2004)